

6 **(long)** A supercharger compressor with an intercooler is fitted to a reciprocating engine as shown in Fig. 6. The air flow conditions at the inlet to the compressor are $T_1 = 300 \text{ K}$ and $p_1 = 1 \text{ bar}$. The air flow at the compressor exit is fed into an intercooler which reduces the air temperature at constant pressure $p_2 = 2 \text{ bar}$ before entering the engine. The engine can be modelled as a steady flow device with heat input per unit mass of air flow of $q_{3-4} = 1 \text{ MJ kg}^{-1}$, and an exhaust temperature of $T_4 = 900 \text{ K}$ at $p_4 = 1 \text{ bar}$. The volumetric flow rate of air into the engine is limited to $\dot{V} = 0.1 \text{ m}^3 \text{ s}^{-1}$. Assume that the cycle fluid is air which can be treated as a perfect gas with heat capacity ratio $\gamma = 1.4$, gas constant $R = 287 \text{ J kg}^{-1} \text{ K}^{-1}$ and specific heat capacity at constant pressure $c_p = 1.005 \text{ kJ kg}^{-1} \text{ K}^{-1}$.

- Sketch a representative $T-s$ diagram for the total process cycle, indicating states 1-4. [5]
- Calculate the temperature at the compressor exit T_2 and specific work w_{1-2} required by the compressor from the engine output assuming isentropic compression. [5]
- Given that the intercooler has a heat transfer per unit mass of air flow of $q_{2-3} = -60 \text{ kJ kg}^{-1}$, calculate the density of air at the exit of the compressor ρ_2 and at the exit of the intercooler ρ_3 . [5]
- Calculate the output power $\dot{W}_{3-4}$ of the engine with and without the intercooler. [5]
- Calculate the total system efficiency of the engine including the compressor and intercooler, and the engine efficiency without the compressor and intercooler, i.e. the engine only. [5]
- Discuss the impact of the compressor and intercooler on the system power and efficiency. [5]

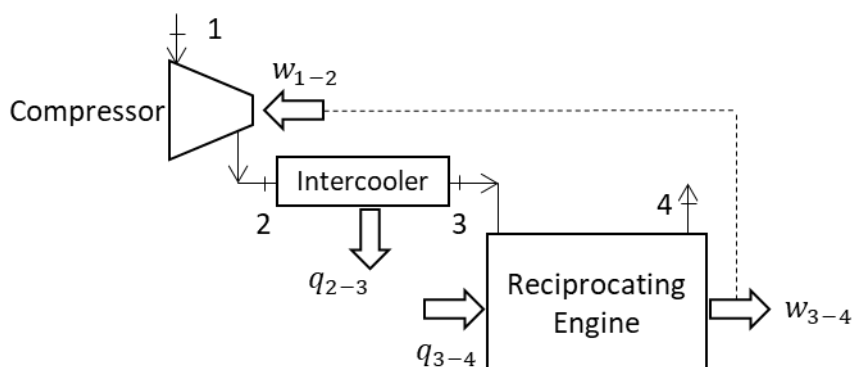


Fig. 6